

lab2

April 3, 2026

1 541 Lab 2 - Privacy

1.1 Required readings

We covered some of these videos during lecture, but I'm including them here so that you have everything in one place. There might be questions on specifics from the required readings whereas the optional readings are more of a general help, optional questions, and if you are interested to explore further.

- **Lec 3** Madhumita Murgia [“How data brokers sold my identity”](#) (0:00 - 9:40)
- **Lec 3** Michal Kosinski [“Part One: The End of Privacy, Data Scientists Know All Your Secrets”](#) (13 min video)
- **Lec 3** Capture Behavioral Engagement [“Marketing Automation for Higher Education”](#) (2 min video)
- Sara Buhr [“An Amazon Echo may be the key to solving a murder case”](#) (3 min read)
- Emma Wollacott [“70,000 OkCupid Profiles Leaked, Intimate Details And All”](#) (4 min read)
- CPAC New clip [“Clearview AI violated Canada’s federal and provincial laws”](#) (5 min video)

1.2 Optional readings

Click to show

You don't have to read these before lab and you don't have to read all of them. The main reason for including them here is as a recommendation of where to deepen your knowledge if there is something that you find particularly interesting. I am giving a brief background to each so you can pick the ones that peak your interest.

- Minute Physics [Protecting Privacy with MATH](#)
 - Made when the US census switched to using differential privacy, includes more examples and intuition behind the concepts.
- Damien Desfontaines [“Differential privacy in \(a bit\) more detail”](#)
 - A more technical explanation of differential privacy, including converting the coin flip example into the equations used when assessing privacy loss.
- Statistics Canada [“A Brief Survey of Privacy Preserving Technologies”](#)
 - Techniques that Canadian government is considering to use to protect your data better.
- Cynthia Dwork [“The Definition of Differential Privacy”](#)
- Chris Gilliard [“Caught in the spotlight”](#)
- Richard Lai [“WiFi mesh networks can detect your breathing”](#)
 - [Fiction short story from Ken Liu on the same topic available here](#) (I really recommend short stories from this author!)

- Glenn Greenwald “[Why privacy matters](#)”
- Rachel Thomas “[Privacy and surveillance](#)”

1.3 Submission instructions

rubric={mechanics:20}

You receive marks for submitting your lab correctly, please follow these instructions:

Follow the general lab instructions.

Click [here](#) to view a description of the rubrics used to grade the questions

Push your .ipynb file to your GitHub repository for this lab (make at least three commits).

Upload your .ipynb file to Gradescope.

Include a clickable link to your GitHub repo for the lab just below this cell

It should look something like this https://github.ubc.ca/MDS-2022-23/DSCI_541_labX_yourcwl.

If you are working in a group, you can create you own (public) repo in the UBC-MDS organization and link that instead.

All your written answers must be in your own words.

You are not allowed to use generative AI tools to write your answers for you or simply paraphrase answer that you generate from these tools (that will lead to a failing grade), but you can use them to further understand the topics you are learning about.

Repo Link : https://github.com/UBC-MDS/DSCI_541_Lab2_group_CodingWithAthul

1.4 Overall writing quality

rubric={writing:20}

You will receive an overall writing grade for the entire lab instead of for each question. This is just a small part of your total grade, but please use the Jupyter Lab spell checker extension to catch typos and read through your text for grammatical errors before submitting (or paste it into Google Docs/MS Word/Grammarly. You don't need to type anything under this cell, it is just a placeholder to generate the grading rubric.

2 1. Short answer questions

Keep your replies brief, 1-3 sentences per question. Although these are short answer questions, don't copy answers from the readings, use your own words so that you practice learning these concepts. These will not be discussed during the lab.

2.1 Question 1.1

rubric={reasoning:60}

What is a data broker?

What are direct and indirect identifiers?

What is k-anonymity and l-diversity?

What are some weaknesses of k-anonymity (and l-diversity)? How could you re-identify individuals in these datasets?

If you have two differentially private datasets, one with and one without your data, what does differential privacy guarantee regarding your privacy?

How does differential privacy work on a conceptual level (watching the optional video from Minute Physics can help with this if you want more details and examples than what was given in the lecture)?

What are some additional approaches you could take to secure data stored in your organization?

YOUR ANSWERS HERE

1. A data broker is basically a company that collects personal info about people from a bunch of different sources like public records, purchase history, and online activity, then sells or licenses that data to other businesses. They kind of act as middlemen in the personal data economy, building up detailed profiles on people who usually don't even know their information is being bought and sold.
2. Direct identifiers are things that can identify a person on their own, like a full name, social security number, or email address. Indirect identifiers (or quasi-identifiers) are things like zip code, date of birth, and gender that don't identify anyone alone but when combined can single out a specific individual pretty easily.
3. K-anonymity is a privacy technique where each record in a dataset is made indistinguishable from at least k-1 other records based on their quasi-identifiers, meaning any given combination of those attributes applies to at least k people. L-diversity builds on this by also requiring that within each k-anonymous group, the sensitive attribute has at least l different values so you still can't determine someone's sensitive info even if you narrow them down to a group.
4. K-anonymity is vulnerable to homogeneity attacks where all records in a group share the same sensitive value, making that value obvious even without identifying the individual. It's also susceptible to background knowledge and linkage attacks where an attacker uses outside information or cross-references with other datasets to re-identify someone, and l-diversity doesn't fully solve this either when the diverse values are semantically similar.
5. Differential privacy guarantees that the output of a query on the dataset would look essentially the same whether your personal data was included or not. So an attacker looking at the results cannot determine with any meaningful confidence whether a specific individual's data was part of the dataset, meaning your individual privacy is protected even if the query results are made public.
6. Differential privacy works by adding carefully calibrated random noise to the results of a query before releasing them. The noise is large enough to mask any single individual's contribution, but small enough that the overall aggregate trends in the dataset are still preserved. A simple way to think about it is the randomized response technique: you ask people a sensitive yes/no question, but have them flip a coin first and answer truthfully only if it lands heads, otherwise they give a random answer. This way you can still estimate the true proportion from the aggregate, but no individual response can be traced back to the truth.

7. Some additional approaches include encrypting data at rest and in transit, enforcing strong access controls with least privilege, and requiring multi factor authentication for sensitive systems. We could also strengthen security through regular backups, with copies stored securely and tested through recovery drills. Network segmentation helps limit how far an attacker can move if one system is compromised. Data loss prevention tools can reduce the risk of accidental or unauthorized sharing of sensitive information. Beyond technical controls, organizations can improve security by classifying data based on sensitivity, applying stricter protections to high risk information, and maintaining detailed audit logs for monitoring and incident investigation.

3 2. Discussion questions

This section asks you to expand a bit on your reasoning, but still aim to write succinct replies around one paragraph per sub-question. The goal of lab discussions are not to provide you with the right answers, but to help your discussion along. Your TA will assist in this by bringing up topics that you might not have thought of, ask questions to break the silence or a dead end, and move the conversation along so that you have time to go through most questions. How useful the lab discussion is for your submission ultimately relies on that you actively contribute to the discussion and help your peers contribute and exchange ideas.

3.1 Some tips to make your discussions in lab more effective

It is easy to overlook the flaws of our own reasoning, so having a discussion with colleagues is an excellent opportunity to develop your thinking and receive feedback from someone who can provide an alternative perspective from your own. Nevertheless, many people don't know how to have an effective discussion, so I am sharing a few tips for you to be able to make the most out of this opportunity:

- Commit to learning, not “winning” debates.
- Comment in order to share information and develop arguments further, not to persuade.
- Listen respectfully, without interrupting, to try to understand each others' views.
 - Don't focus on what you are going to say next while someone else is talking.
- Challenge ideas, not individuals.
 - And be open to having your own ideas challenged.
- Think about as good arguments as possible against your position.
 - This is especially useful if many of your peers have the same opinion, help your group find angles that you might otherwise be missing.
- Allow everyone the chance to speak.
 - Politely ask members of your group about their opinion.
- Avoid assumptions about any member of the class or generalizations about social groups.
 - Be careful about asking individuals to speak on the behalf of their (perceived) social group.
- Be aware of [logical fallacies](#), but avoid pointing them out in rude or disrespectful ways.

3.2 Question 2.1 – Technology and privacy

rubric={reasoning:100}

Acceptable exchange. Do you consider it acceptable that companies collect information about

your online behavior in exchange for using their services? Is the service they provide a reasonable compensation of your data or do you think they owe you monetary compensation if they profit from selling your data to a third party? Include an example of a service you think provides a reasonable or unreasonable trade off.

Pay for privacy. If you had the option would you pay to use email, social media, file storage, etc that met your preferred privacy standard? Are there specific online services which you would be more likely to pay for than others (and why do you think privacy is more important in these cases)? Is privacy a commodity that should only be available to those who can afford it, or is it a right that should be regulated and available to all for free?

OUTLINE SUBMITTED BY THE END OF THE LAB (if applicable)

1. It's acceptable in the general sense, but currently most of the Social media platforms have normalized using user data not just for improving their service, but for monetization the data in itself, which is not acceptable.
2. Freedom to choose for Paying for privacy should be there, but realistically speaking there are a lot of problems from paying for privacy. Privacy related data can be used for any kind of purposes now or in the future. eg: ProtonVPN allows you to use VPN service for free, but they can access your data, and no one really knows what really is done with the data.

FINAL OUTLINE

1. In a general sense, using personal data to improve platforms is acceptable. However, most social media platforms have gone beyond that and normalized monetizing user data itself, not just using it to enhance their service. That crosses the line from acceptable to exploitative.
2. Users should have the freedom to pay for privacy if they choose to, but in practice there are real problems with this model. Privacy-related data can be repurposed for any number of things now or in the future. For example, ProtonVPN offers a free tier, but in exchange they can access your usage data, and there is no real transparency about what is actually done with it.

YOUR ANSWERS HERE

1. Using personal data to improve a platform is kind of acceptable in a general sense, like if a service genuinely gets better because of my data thats a somewhat fair trade off we think. But the problem is that most social media platforms have gone way past that point and basically normalized treating user data as its own separate revenue stream, completely disconnected from any real improvement to the service they actually provide. That shift from using data to make the product better to selling the data as the product itself is where it crosses from acceptable to exploitative in my opinion, and its a distinction that I feel tends to get lost a lot in these conversations.
2. The freedom to pay for privacy should exist in principle, but in practice theres genuinely serious problems with how that model actually works out. Even services marketed as privacy focused can be pretty misleading, ProtonVPN is a good example where the free tier sounds great but in exchange they can access your usage data and theres basically no real transparency about what actually gets done with it or when. This shows that privacy related data can be repurposed for all kinds of things at any point really, and the assumption that paying for something automatically buys you actual privacy protection is not nearly as solid

as people tend to think, which is kind of a concerning realization when you start looking into it.

3.3 Question 2.2 – Technology and privacy

rubric={reasoning:100}

Terms of service. When we sign up for online services, we often have to agree to terms of conditions to use the services. Sometimes these do indeed mention that your data is being collected by the company, but studies have shown that the language used in terms and conditions often requires college level reading apprehension, while many users of such platforms are children in middle and high school. Could you think of some ways to improve the presentation of terms of services to convey the information more effectively?

Alternative options. A common argument is that we can just opt out of services we don't like and use others instead (i.e. it is the individual's responsibility), but is this really true? Or are we getting to the stage where we "need" certain online services for us to integrate in society and opting out is not really an option? For example, is there a widely used replacement alternative for services such as Facebook, Instagram, YouTube or TikTok that does not center their business model on mining user information?

Beyond online tracking. What about extending the data collection outside the online environment? Which of the following do you find reasonable and unreasonable privacy-wise? Are these examples different from tracking online (why/why not)?

Is it ok if an Amazon Echo/Google Home device records your voice at home?

What if you have friends over, do you need to inform them about your echo device and should their voices be recorded too?

What about a pair of glasses that automatically record video and sound as you walk around in public places?

Or using WiFi signals to detect people moving in their homes?

OUTLINE SUBMITTED BY THE END OF THE LAB (if applicable)

1. Adding friction to TOS to make users pay more attention to it, making it simple to read (perhaps add some illustrations or something) by adding regulations that the company need to abide by so that companies can't abuse the 'Easy TOS accept interface' to make users easily accept TOS without actually reading it.
2. No, the alternative options argument isn't good because then all alternatives will use the complex TOS system, then user won't be able to find any alternative application for their purpose. (It is apparent in most social media platforms) and those that have easier or acceptable TOS, aren't popular services which puts the user on the back step.
3. Tracking data beyond online should be very transparent, like Amazon Echo TOS doesn't quite explicitly say that its also listening to 'non-command' audio and collecting data from it. There should be guardrails in place not collect 'non-command' audio from users. It should be an absolute NO to beyond-online tracking.

FINAL OUTLINE

1. One approach is to add friction to the Terms of Service process so that users actually pay attention before accepting. This could include making the language simpler, adding illustrations, or introducing regulations that prevent companies from abusing the “easy accept” interface to push users through without reading anything.
2. No, the “just use an alternative” argument does not hold up. If all major platforms use the same kind of complex TOS, there is no meaningful alternative for users to switch to. This is already apparent across most social media platforms. The few services that do have simpler or more reasonable TOS tend to be less popular, which puts users at a disadvantage since they lose access to the network effects of the larger platforms.
3. Tracking data beyond online activity should be handled with full transparency. For instance, Amazon Echo’s TOS does not explicitly state that it also listens to non-command audio and collects data from it. There should be strict guardrails in place to prevent collecting non-command audio from users. Beyond-online tracking should be an absolute no.

YOUR ANSWERS HERE

1. The way TOS acceptance is designed right now basically guarantees nobody reads it. You scroll past a wall of legal text, hit “I agree,” and move on. One way to fix this is to make companies actually work for that click. Require a plain-language summary upfront, maybe a simple graphic showing what data gets collected and who sees it, and some kind of friction before acceptance like a mandatory scroll. The regulation angle matters here too, because without it companies have zero incentive to change anything. If anything, a frictionless accept button benefits them.
2. I don’t think the “just use an alternative” argument holds up. Try naming a real alternative to YouTube or Instagram that doesn’t run on the same data model. It’s hard. The platforms that do have cleaner terms are usually small enough that they’re not real substitutes, especially for anything social where the whole point is that other people are on it. So in practice the choice is between the big platforms on their terms, or not using them at all. That’s not much of a choice, and framing it as individual responsibility kind of ignores that reality.
3. Beyond-online tracking is where the bar should be much higher than it currently is. Amazon Echo is a good example of the problem. Most people assume it only picks up audio after the wake word, but that’s not quite how it works, and the TOS doesn’t make that clear. There’s a real difference between a website tracking which pages you visit and a device sitting in your kitchen picking up audio. One you can close, the other you’d have to physically unplug. Non-command audio collection should just not be allowed. Not disclosed in fine print, actually prohibited.

3.4 Question 2.3 – Scraping data and facial recognition software

rubric={reasoning:100}

Facilitating queries on scraped data. Is it fair game to scrape public information from the internet, and then facilitate queries on that information or are there any additional concerns when the data becomes easier to access? What about scraping data that is only public for anyone logged in to that platform but otherwise not accessible (e.g. Facebook or OKCupid) and then making it available outside the platform?

Privacy and control of scraped data. Clearview AI is a company that provides facial recognition

software, which is used by private companies, law enforcement agencies, universities and individuals. The database consists of more than three billion images scraped from the internet, including from social media applications. Clearview claims 99% accuracy for most photos and one of their goals is to find criminals more accurately than current approaches used by law enforcement. One of its biggest technical advantages over previous law enforcement tools is that they have a huge set of images including the general public, not just photos of previous convicts. A recent data breach revealed that Clearview AI is employed among American and Canadian law enforcement, including by the Vancouver Police Department.

How do you think such powerful facial recognition technology should be regulated? For example, is it fair use to scrape data that was public at the time of scraping and store it permanently in a database or do we have a right to be forgotten even in downloaded data? Should we take into consideration that they are using this data for a good cause, such as catching criminals (e.g. Clearview is claimed to have been used to identify participants in the Capitol hill riot and to identify dead soldiers in the Russia/Ukraine war)? Also think about how the scale of implementation matters, even for algorithms that are 99% accurate.

OUTLINE SUBMITTED BY THE END OF THE LAB (if applicable)

1. Say, If your facebook account is public anyway, so we think its fine, as the data from that is easy to access anyway with just account creation on the platform, so yes its fair game to scrape already public information on the internet.
2. So its better to use AI technology to find such criminals because it saves valuable resources , and its for the public good, while ensuring proper guardrails and regulatory bodies have control over this.

FINAL OUTLINE

1. If someone's Facebook profile is already public, then scraping that data is fair game. The information is easy to access anyway with just an account on the platform, so scraping publicly available data from the internet is acceptable.
2. Using AI technology to identify criminals is a net positive because it saves valuable investigative resources and serves the public good, as long as there are proper guardrails in place and regulatory bodies maintain oversight over how the technology is used.

YOUR ANSWERS HERE

1. If someone's profile information is already publicly accessible, scraping that data is generally fair game since anyone can see it with a quick search or by just creating an account. However, the concern comes when you take that scattered public data and aggregate it into a searchable database, because that fundamentally changes how easy it is to access. A photo sitting on someone's public Facebook is one thing, but indexing it alongside millions of other photos and making it queryable changes the privacy dynamic entirely. For data that is only accessible to logged-in users of a platform, scraping and redistributing that outside the platform clearly violates the implicit expectation of the user who shared it within that specific context. The OkCupid leak is a good example where intimate details were exposed to anyone outside the platform without users' consent.
2. Using AI for identifying criminals can be a net positive because it saves valuable investigative resources and serves the public good, and Clearview's use in cases like the Capitol Hill riot shows it has real utility. That said, scraping billions of images from the internet without

consent and storing them permanently raises serious concerns about the right to be forgotten. Even at 99% accuracy, at scale that 1% error translates to millions of false matches, which could lead to innocent people being flagged or investigated. So the technology should only be available to law enforcement under judicial oversight, not to private companies or individuals, and there need to be strict regulations around data retention and the ability for people to request deletion of their data from these databases.

3.5 Question 2.4 (Challenging)

rubric={reasoning:20}

Note: Since this exercise is a bit longer than the regular challenging questions, it will count both for the ~5% challenging points for this lab, and the 2% extra credit assignment for your overall grade in 541.

In previous years, students have asked for an exercise on how to implement differential privacy in a data analysis. It's difficult to fit this into the curriculum because many of the differential privacy libraries are not trivial to use and would take a fair amount of time to learn properly.

This challenging exercise is an experiment collaborated with the team behind Antigranular, a community-led open-source platform that combines confidential computing with differential privacy. They have put together a notebook with an introduction to a couple of differential privacy libraries that are easier to get started with. Antigranular is aiming to develop a general curriculum for privacy concerned data analysis, so as part of this exercise you will give feedback on how easy it was to follow along in the tutorial notebook and what you think could be improved.

You can access the notebook in the student repo. You can either study the output of the notebook without running it, or re-run the cells and change things around if you want to do more experimentation. Your task is to read through the notebook and answer the following questions:

Why is the salary that is printed for the new employee so different when using differential privacy versus when we are not using it? Would you say this is a feature of differential privacy or a drawback?

What does it mean that there is a tradeoff between privacy and utility? Describe this in the context of differential privacy specifically referring to the figure with the same name in the notebook and comment on why the size of the data matters.

Provide feedback on your general experience working through the notebook. What worked well and what do you think could be improved? Was it suitable for someone with your background level of experience with differential privacy (a brief high level introduction to the main concepts during lecture)? Did you find that seeing examples in actual code helpful for your understanding of what differential privacy is and how it can be employed in a data analysis?

YOUR ANSWERS HERE

1. Differential privacy produces a very different salary for the new employee because it adds random noise to the output in order to protect that individual's information. Without differential privacy, the result reflects the true value much more directly. However, with differential privacy, the answer is intentionally distorted so that it becomes harder to infer one person's exact data. In this case, since the result is strongly influenced by a single employee's salary, the noisy output can look very different from the non-private one. We observe that this

is mainly a feature of differential privacy, because the difference is exactly what helps preserve privacy. However, it also shows the practical downside that the result may become less accurate.

2. The tradeoff between privacy and utility means that stronger privacy protection usually reduces the accuracy or usefulness of the results. In differential privacy, this happens because stronger privacy requires adding more noise. The figure in the notebook illustrates that as privacy increases, utility tends to decrease, and as utility increases, privacy tends to weaken. The size of the data matters because noise affects small datasets more strongly than large datasets. In a large dataset, the overall pattern is more stable, so the added noise has less impact on the final result. In a small dataset, each individual contributes more to the outcome, so the noise can change the answer much more noticeably.
 3. Overall, we found the notebook to be a helpful introduction to differential privacy. Since my background is mainly a brief high level introduction from lecture, the notebook felt accessible and appropriate. What worked well was that it combined explanation with actual code examples, which made the ideas easier to understand in practice. Seeing the code helped us better understand that differential privacy is not just about hiding data, but about changing computations in a structured way to protect individuals. One thing that could be improved is adding a bit more explanation after some outputs, especially for beginners, so it is clearer what each result demonstrates. In general, though, I found the notebook useful, and the hands-on examples definitely improved my understanding of how differential privacy can be used in data analysis.
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4 Help us improve the labs

The MDS program is continually looking to improve our courses, including lab questions and content. The following optional questions will not affect your grade in any way nor will they be used for anything other than program improvement:

1. Approximately how many hours did you spend working or thinking about this assignment (including lab time)?

#Ans: 3

2. Were there any questions that you particularly liked or disliked?

#Ans: [Questions you liked] : Ex 2.2

#Ans: [Questions you disliked] : Ex 2.4